

CLASS 11-CBSE MATHEMATICS

CHAPTER: PROBABILITY-SET-1

TIME : 1Hr

Mx.Marks:30

SECTION A

Q.I Answer the following questions

[1X3 = 3 MARKS]

1. Three of the six vertices of a regular hexagon are chosen at random. What is the probability that the triangle with these three vertices is equilateral?
2. A box contains 6 nails and 10 nuts. Half of the nails and half of the nuts are rusted. If one item is chosen at random, what is the probability that it is rusted or is a nail.
3. A box contains 1 red and 3 identical white balls. Two balls are drawn at random in succession without replacement. Write the sample space for this experiment.

SECTION B

Q.II Answer the following questions

[2X5 = 10 MARKS]

4. Two integers are chosen at random and multiplied. What is the probability that the product is an even integer?
5. If $(1+3p)/3$, $(1-p)/4$ and $(1-2p)/2$ are the probabilities of three mutually exclusive events, then find the set of all values of p
6. A card is chosen randomly from a pack of playing cards. Find the probability that it is a black king or queen of heart or jack.
7. A fair coin with 1 marked on one face and 6 on the other and a fair die are both tossed. Find the probability that the sum of numbers that turn up is (i) 3 (ii) 12
8. A and B are events such that $P(A) = 0.42$, $P(B) = 0.48$ and $P(A \text{ and } B) = 0.16$. Determine (i) $P(\text{not } A)$, (ii) $P(\text{not } B)$ and (iii) $P(A \text{ or } B)$.

SECTION C



Q.III Answer the following questions

[3X3 = 9 MARKS]

9. Two dice are thrown. The events A, B and C are as follows:

A: getting an even number on the first die.

B: getting an odd number on the first die.

C: getting the sum of the numbers on the dice ≤ 5 .

Describe the events

(i) A' (ii) not B (iii) A or B

10. A pack of cards contains 4 aces, 4 kings, 4 queens and 4 jacks. Two cards are drawn at random. Find the probability that at least one of them is an ace.

11. Without repetition of the numbers, four-digit numbers are formed with the numbers 0, 2, 3, 5. Find the probability of such a number divisible by 5.

SECTION D

Q.IV Answer the following questions

[4X2 = 8 MARKS]

12. Out of 30 consecutive integers, 2 are chosen at random. Find the probability that their sum is odd.

13. Three-digit numbers are formed using the digits 0, 2, 4, 6, 8. A number is chosen at random out of these numbers. What is the probability that this number has the same digits?

